

Polynomials, factor and remainder theorems - Set 2 - Solutions

Specification MM1.6.

Answer key

Question	Answer	Topic
1	H	Polynomials, factor and remainder theorems
2	B	Polynomials, factor and remainder theorems
3	A	Polynomials, factor and remainder theorems
4	B	Polynomials, factor and remainder theorems
5	F	Polynomials, factor and remainder theorems

Worked solutions

Question 1 POL-08 · Polynomials, factor and remainder theorems · spec MM1.6

Which of the following statements about the polynomial $p(x) = x^3 - 6x^2 + 11x - 6$ is/are true?

I $p(x)$ has three distinct real roots.

II $(x - 4)$ is a factor of $p(x) - 6$.

III $p(x) > 0$ for all $x > 3$.

Answer: H

$p(x) = (x - 1)(x - 2)(x - 3)$, so I is true. $p(4) - 6 = 3 \cdot 2 \cdot 1 - 6 = 0$, so II is true. For $x > 3$ all three factors are positive, so III is true.

Full method - how and why

Three statements about one cubic. Factorising it first makes all three immediate.

1 Factorise: $p(1) = 1 - 6 + 11 - 6 = 0$, so $(x - 1)$ is a factor and $p(x) = (x - 1)(x - 2)(x - 3)$.

why: Test the divisors of 6; the resulting quadratic $x^2 - 5x + 6$ factorises easily.

2 I: the roots 1, 2, 3 are distinct. **True.**

why: Three distinct linear factors means three distinct roots.

3 II: $(x - 4)$ is a factor of $p(x) - 6$ iff $p(4) - 6 = 0$. Now $p(4) = 3 \times 2 \times 1 = 6$, so $p(4) - 6 = 0$. **True.**

why: Factor Theorem applied to the shifted polynomial - evaluate the whole expression $p(x) - 6$ at $x = 4$.

4 III: for $x > 3$ each factor $x - 1$, $x - 2$, $x - 3$ is positive, so the product is positive. **True.**

why: Sign analysis from the factorised form; beyond the largest root everything is positive.

5 All three: 'I, II and III'.

why: Match verdicts to options.

Where it goes wrong

- Expanding $p(x) - 6$ and re-testing from scratch instead of just evaluating $p(4) - 6$.
- Checking III at one value like $x = 10$ and calling it proved - the factor argument covers every $x > 3$.

Takeaway Factorise first: roots, sign and shifted-factor questions then all read off the same form.

Question 2 POL-09 · Polynomials, factor and remainder theorems · spec MM1.6

The expression $x^4 + 4$ can be factorised into two quadratics with integer coefficients. What is the sum of the coefficients of x in these two quadratics?

Answer: B

$x^4 + 4 = (x^2 + 2)^2 - 4x^2 = (x^2 + 2x + 2)(x^2 - 2x + 2)$. The x -coefficients are 2 and -2 , summing to 0.

Full method - how and why

$x^4 + 4$ has no real roots, so it cannot be split into linear factors - but it can be split into two quadratics via a difference of two squares.

1 Complete the square in x^2 : $x^4 + 4 = (x^2 + 2)^2 - 4x^2$.

why: $(x^2 + 2)^2 = x^4 + 4x^2 + 4$, which is $4x^2$ too big - so subtract it back. This manufactured square is the standard trick (the Sophie Germain identity).

2 Recognise a difference of two squares: $A^2 - B^2$ with $A = x^2 + 2$, $B = 2x$.

why: $4x^2 = (2x)^2$, so the expression is now in the form that factorises.

3 Factorise: $(x^2 + 2 - 2x)(x^2 + 2 + 2x) = (x^2 - 2x + 2)(x^2 + 2x + 2)$.

why: $A^2 - B^2 = (A - B)(A + B)$.

4 Add the x -coefficients: $-2 + 2 = 0$.

why: The question asks only for this sum, which the symmetry makes obvious.

Where it goes wrong

- Concluding it cannot be factorised because it has no real roots - a quartic with no real roots still factorises into two quadratics.
- Adding the constant terms ($2 + 2 = 4$) instead of the x -coefficients.

Takeaway $x^4 + 4a^4 = (x^2 - 2ax + 2a^2)(x^2 + 2ax + 2a^2)$: manufacture a square, then use the difference of two squares.

Question 3 POL-10 · Polynomials, factor and remainder theorems · spec MM1.6

Given that $x^3 + px + q$ is divisible by $(x - 2)^2$, find the value of q .

Answer: A

$x^3 + px + q = (x - 2)^2(x - r)$. Comparing x^2 coefficients: $-4 - r = 0 \Rightarrow r = -4$. Constant term: $(4)(4) = 16 = q$.
(Also $p = 4 + 4r = -12$.)

Full method - how and why

A repeated factor $(x - 2)^2$ means $x = 2$ is a double root. Comparing coefficients in the factorised form gives q without any calculus.

1 Write $x^3 + px + q = (x - 2)^2(x - r)$.

why: A cubic with a repeated root 2 and one other root r must take this form (leading coefficient 1).

2 Compare x^2 coefficients. The left-hand side has none, so the coefficient is 0; expanding the right gives $-(4 + r)$... more precisely $(x^2 - 4x + 4)(x - r)$ has x^2 coefficient $-r - 4$.

why: The **absence** of a term is information: its coefficient is zero.

3 So $-r - 4 = 0$, giving $r = -4$.

why: One comparison is enough to find the third root.

4 Constant term: $4 \times (-r) = 4 \times 4 = 16$, so $q = 16$.

why: The constant of $(x - 2)^2(x - r)$ is $4 \times (-r)$; with $r = -4$ that is 16.

Where it goes wrong

- Forgetting that the missing x^2 term forces a condition, and trying to solve with only the constant term.
- Sign slip on $-r$, giving $q = -16$.

Takeaway A missing power in a polynomial is a coefficient equal to zero - often the most useful equation available.

Question 4 POL-12 · Polynomials, factor and remainder theorems · spec MM1.6

When $x^3 - 3x^2 + 2x + 7$ is divided by $x^2 + 1$, what is the remainder?

Answer: B

$x^3 - 3x^2 + 2x + 7 = (x^2 + 1)(x - 3) + (x + 10)$: expanding gives $x^3 - 3x^2 + x - 3 + x + 10$. Remainder $x + 10$.

Full method - how and why

Division by a quadratic again: remainder is linear. Comparing coefficients is fastest because the divisor has no x term.

- 1 Write $x^3 - 3x^2 + 2x + 7 = (x^2 + 1)(x + c) + (ax + b)$.
why: Cubic divided by quadratic gives a linear quotient and a linear remainder.
- 2 Expand the right: $x^3 + cx^2 + x + c + ax + b$.
why: $(x^2 + 1)(x + c) = x^3 + cx^2 + x + c$.
- 3 Compare x^2 : $c = -3$.
why: Only the quotient contributes to x^2 .
- 4 Compare x : $1 + a = 2$, so $a = 1$. Compare constants: $c + b = 7$, so $b = 7 + 3 = 10$.
why: Work down the powers; each comparison introduces exactly one new unknown.
- 5 Remainder: $x + 10$.
why: Degree 1 < degree 2 ✓.

Where it goes wrong

- Forgetting the $+x$ that $(x^2 + 1)(x)$ contributes, giving $a = 2$ and the remainder $2x + 10$.
- Using the Remainder Theorem with $x^2 + 1$ - it has no real roots, so that route needs complex numbers and is off-specification.

Takeaway Comparing coefficients handles any divisor; only use root substitution when the divisor has real roots.

Question 5 POL-16 · Polynomials, factor and remainder theorems · spec MM1.6

For which value of k does $x^2 - x + k$ divide exactly into $x^3 - 2x^2 + 3x - 4$? (Assume such a value exists.)

Answer: F

Try $x^3 - 2x^2 + 3x - 4 = (x^2 - x + k)(x + m)$: comparing x^2 : $m - 1 = -2 \Rightarrow m = -1$. Then x : $k - m = k + 1 = 3 \Rightarrow k = 2$. Constant: $km = -2 \neq -4$. No value of k works.

Full method - how and why

An 'assume it exists' question where the honest answer is that it does not. Comparing coefficients over-determines k , and the system is inconsistent.

1 If $x^2 - x + k$ divides exactly, the quotient is linear: $x^3 - 2x^2 + 3x - 4 = (x^2 - x + k)(x + m)$.

why: Cubic $\div$ quadratic = linear, and 'divides exactly' means no remainder term.

2 Compare x^2 : $m - 1 = -2$, so $m = -1$.

why: The x^2 terms are mx^2 and $-x^2$.

3 Compare x : $k - m = 3$, so $k = 2$.

why: The x terms are kx and $-mx$.

4 Check the constant term: $km = 2 \times (-1) = -2$, but the cubic has -4 . Contradiction.

why: With three equations and two unknowns the system is over-determined - the last equation is a genuine test, not a formality.

5 So no value of k works: the answer is 'no such value exists'.

why: The bracketed 'assume such a value exists' is a deliberate misdirection.

Where it goes wrong

- Stopping at $k = 2$ and never checking the constant term.
- Trusting the question's hint that a value exists - always verify all the coefficients.

Takeaway When comparing coefficients gives more equations than unknowns, the leftover equations must be checked; they can reveal that no solution exists.